

Binomial Distribution



Objectives:

- Define binomial distribution and how it is used.
- Demonstrate how binomial distribution is computed.
- Demonstrate binomial distribution with the use of Google Colab

Overview

- For this discussion, we will be using Numpy and Matplotlib.
- Matplotlib is a plotting library for Python. It is used along with NumPy to provide an environment that is an effective open source alternative for MatLab. It can also be used with graphics toolkits like PyQt and wxPython.

Overview

- Matplotlib module was first written by John D. Hunter. Since 2012, Michael Droettboom is the principal developer. Currently, Matplotlib ver. 1.5.1 is the stable version available. The package is available in binary distribution as well as in the source code form on
- Here **pyplot()** is the most important function in matplotlib library, which is used to plot 2D data.

What is a binomial distribution?

- Binomial Distribution is a *Discrete Distribution*.
- It describes the outcome of binary scenarios, e.g. toss of a coin, it will either be head or tails.

What is a binomial distribution?

- Yes/no (binomial) outcomes lie at the heart of analytics since they are often the culmination of a decision or other process; buy/don't buy, click/don't click, survive/die, and so on.
- Central to understanding the binomial distribution is the idea of a set of trials, each trial having two possible outcomes with definite probabilities.

Understanding Binomial Distribution

- The binomial distribution is the frequency distribution of the number of successes (x) in a given number of trials (n) with specified probability (p) of success in each trial.
- There is a family of binomial distributions, depending on the values of n and p .

Binomial Probability

- 1. the experiment has a fixed number of trial
- 2. the trials are independent
- 3. each trial has 2 possible outcomes (ex. Pass/Fail, Head/Tail)
- 4. probability of success is consistent for each trial

$$B(n, p)$$

Where: n- number of trials

P-probability of success

Notations

x =number of success

n = number of trials

p =probability of success

q =probability of failure

It is important to determine whether an experiment is Binomial Probability Distribution or not.

Example 1

- Miriam did not study for an upcoming quiz. Luckily, the quiz is multiple choice type. Miriam decided to guess on a 10-item quiz where each problem has 5 choices. What is the probability that Miriam will get 7 of the questions correctly?

Solution:

- $n=10$
- $x=7$
- $p= 1/5$ or 0.20
- $q=4/5$ or 0.80

Example 2

- What is the probability that Miriam will get 7 or less questions answered correctly by guessing?

Example 2

$$\begin{aligned}
 x &= 6 \\
 nCx \cdot p^x \cdot q^{n-x} &= {}^{10}C_6 \cdot (0.2)^6 \cdot (0.8)^4 \\
 &= 210 \cdot 0.000064 \cdot 0.4096 \\
 &= 0.0055050 \times 100 = \boxed{0.55\%}
 \end{aligned}$$

$$x \approx 0.079\%$$

$$P(X \leq 7)$$

$$\begin{aligned}
 &= P(0) + P(1) + P(2) + P(3) + P(4) + P(5) + P(6) + P(7) \\
 &= 10.73 + 26.84 + 30.186 + \dots \\
 &= \boxed{99.1\%}
 \end{aligned}$$

- What is the probability that Miriam will get 7 or less questions answered correctly by guessing?

Solution:

0	1	2	3	4	5	6	7	8	9	10
							*			

$$n=10$$

$$x = ?$$

$$p = 1/5 \text{ or } 0.20$$

$$q = 4/5 \text{ or } 0.80$$

$$x=0$$

$$\begin{aligned}
 &= nCx \cdot p^x \cdot q^{n-x} \\
 &= {}^{10}C_0 \cdot p^0 \cdot q^{10-0} \\
 &= 1 \cdot (0.20)^0 \cdot q^{10}
 \end{aligned}$$

$$\begin{aligned}
 &1 \cdot 1 \cdot 0.1073 \\
 &= 0.10737 \times 100 \\
 &= \boxed{10.73\%}
 \end{aligned}$$

$$x=1$$

$$\begin{aligned}
 &= nCx \cdot p^x \cdot q^{n-x} \\
 &= {}^{10}C_1 \cdot (0.20)^1 \cdot (0.80)^9 \\
 &= 10 \cdot (0.20) \cdot (0.1342) \\
 &= 0.2684 \text{ or } \boxed{26.84\%}
 \end{aligned}$$

$$x=2$$

$$\begin{aligned}
 &= nCx \cdot p^x \cdot q^{n-x} \\
 &= {}^{10}C_2 \cdot (0.20)^2 \cdot (0.80)^8 \\
 &= 45 \cdot (0.04) \cdot 0.1677 \\
 &= 0.30186 \text{ or } \\
 &= \boxed{30.186\%}
 \end{aligned}$$

$$x=3$$

$$\begin{aligned}
 &= nCx \cdot p^x \cdot q^{n-x} \\
 &= {}^{10}C_3 \cdot (0.20)^3 \cdot (0.80)^7 \\
 &= 120 \cdot 0.008 \cdot 0.2097 \\
 &= 0.20131 = \boxed{20.13\%}
 \end{aligned}$$

$$x=4$$

$$\begin{aligned}
 &= nCx \cdot p^x \cdot q^{n-x} \\
 &= {}^{10}C_4 \cdot (0.20)^4 \cdot (0.80)^6 \\
 &= 210 \cdot 0.0016 \cdot (0.262) \\
 &= 0.088032 \times 100 = \boxed{8.8032\%}
 \end{aligned}$$

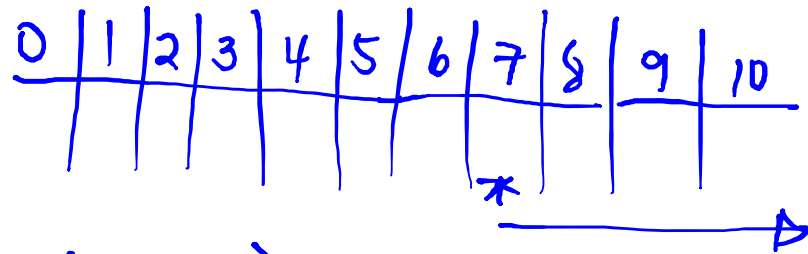
$$x=5$$

$$\begin{aligned}
 &= nCx \cdot p^x \cdot q^{n-x} \\
 &= {}^{10}C_5 \cdot (0.20)^5 \cdot (0.80)^5 \\
 &= 252 \cdot 0.00032 \cdot 0.32768 \\
 &= 0.02642 \text{ or } \boxed{2.64\%}
 \end{aligned}$$

Example 3

- What is the probability that Miriam will get 7 or more correct answers?

- Solution:



$$P(X \geq 7)$$

$$= 1 - \text{sum of probability of } \underline{7, 8, 9, 10}$$

$$=$$

$$\underline{\hspace{2cm}}$$

Example 4

You are rolling a 6-sided die 12 times. You want to find out the probability of getting 3 as an outcome 5 times.

Solution:

Example 5

- When an unbiased coin is tossed 8 times, what is the probability of obtaining
 - a. Less than 4 heads
 - b. More than 5 heads?

Your turn

Create and plot binomial
distribution in Python

We will first import the required modules:

- Note: Basis for this demonstration is data from example 1
- Step 1: import required modules

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom
```

- Step 2: Create values in Python
- Step 3: Construct Probability Mass Function
- Step 4: Create visualization

```
#creating values in Python
n=10
p=0.20
x=np.arange(0,n+1)

#construct Probability Mass Function (PMF)
binomial_pmf=binom.pmf(x, n, p)
print (binomial_pmf)

#visualization
plt.plot(x, binomial_pmf, color='blue')
plt.title(f"Binomial Distribution(n={n}, p={p})")
plt.show()
```

Output for Example 1

Example 1: Miriam did not study for an upcoming quiz. Luckily, the quiz is multiple choice type. Miriam decided to guess on a 10-item quiz where each problem has 5 choices. What is the probability that Miriam will get 7 of the questions correctly?

```
[ ] #Example 1:
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom

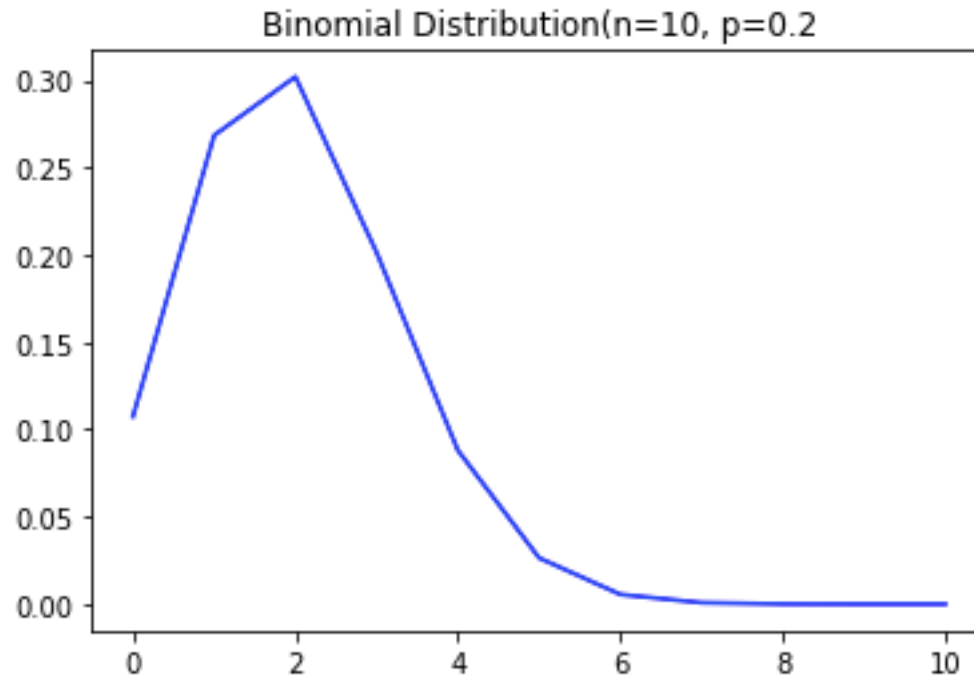
#creating values in Python
n=10
p=0.20
x=np.arange(0,n+1)

#construct Probability Mass Function (PMF)
binomial_pmf=binom.pmf(x, n, p)
print (binomial_pmf)

#visualization
plt.plot(x, binomial_pmf, color='blue')
plt.title(f"Binomial Distribution(n={n}, p={p})")
plt.show()
```

Output for Example 1

```
[1.07374182e-01 2.68435456e-01 3.01989888e-01 2.01326592e-01  
8.80803840e-02 2.64241152e-02 5.50502400e-03 7.86432000e-04  
7.37280000e-05 4.09600000e-06 1.02400000e-07]
```



From Example 4:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom

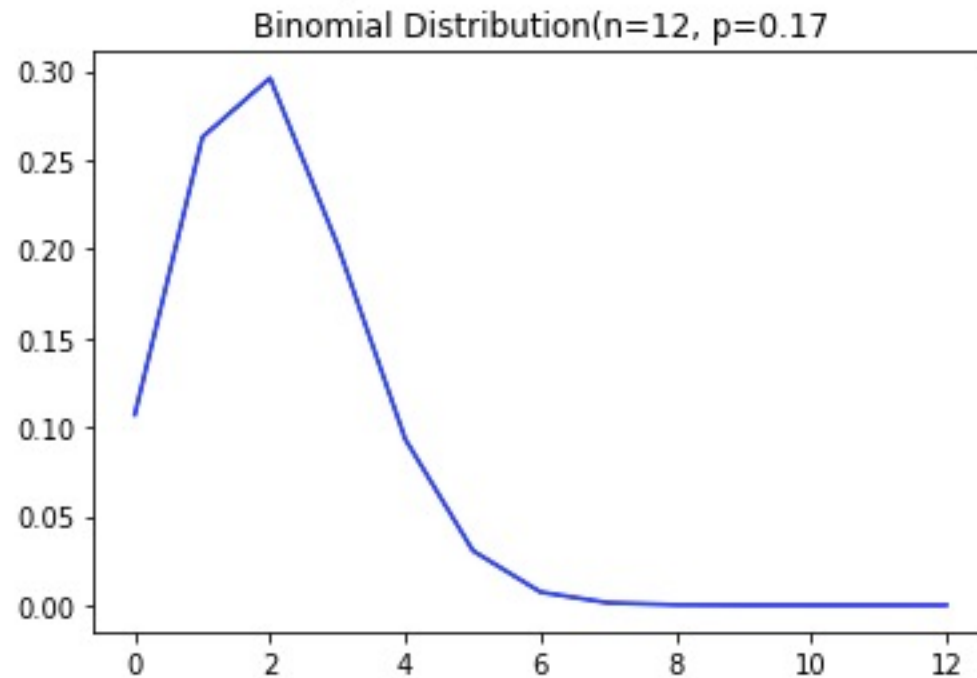
#creating values in Python
n=12
p=0.17
x=np.arange(0,n+1)

#construct Probability Mass Function (PMF)
binomial_pmf=binom.pmf(x, n, p)
print (binomial_pmf)

#visualization
plt.plot(x, binomial_pmf, color='blue')
plt.title(f"Binomial Distribution(n={n}, p={p}")
plt.show()
```

Output of Example 4

```
[1.06890008e-01 2.62717609e-01 2.95952970e-01 2.02056244e-01  
9.31162813e-02 3.05152151e-02 7.29178834e-03 1.28014184e-03  
1.63873579e-04 1.49175414e-05 9.16620011e-07 3.41348087e-08  
5.82622237e-10]
```



The Shape of Binomial Distribution

- The shape or pattern of binomial distribution depends on the value of p and n .
- If $p=q=0.5$, the distribution will be symmetrical regardless of the values of n .
- If p is not equal to q , the distribution will be asymmetrical.
- Given a particular n , the more the difference between p & q , the greater the skewness of the distribution will be.

The Shape of Binomial Distribution

- When $p < q$, the distribution will be positively skewed.
- When $p > q$, then the distribution will be negatively skewed.
- When n becomes infinitely large, the distribution will approach symmetry irrespective of the difference between p and q .

Real Life Examples of the Binomial Distribution

- The Binomial distribution is a probability distribution that is used to model the probability that a certain number of “successes” occur during a certain number of trials.

Real Life Examples of the Binomial Distribution

- **Example 1: Number of Side Effects from Medications**
- Medical professionals use the binomial distribution to model the probability that a certain number of patients will experience side effects as a result of taking new medications.
- For example, suppose it is known that 5% of adults who take a certain medication experience negative side effects.

Real Life Examples of the Binomial Distribution

- We can use a Binomial Probability to find the probability that more than a certain number of patients in a random sample of 100 will experience negative side effects.
- $P(X > 5 \text{ patients experience side effects}) = \mathbf{0.38400}$
- $P(X > 10 \text{ patients experience side effects}) = \mathbf{0.01147}$
- $P(X > 15 \text{ patients experience side effects}) = \mathbf{0.0004}$

And so on.

- This gives medical professionals an idea of how likely it is that more than a certain number of patients will experience negative side effects.

Real Life Examples of the Binomial Distribution

- **Example 2: Number of Fraudulent Transactions**
- Banks use the binomial distribution to model the probability that a certain number of credit card transactions are fraudulent.
- For example, suppose it is known that 2% of all credit card transactions in a certain region are fraudulent.

Real Life Examples of the Binomial Distribution

- If there are 50 transactions per day in a certain region, we can use Binomial Probability to find the probability that more than a certain number of fraudulent transactions occur in a given day:
- $P(X > 1 \text{ fraudulent transaction}) = \mathbf{0.26423}$
- $P(X > 2 \text{ fraudulent transactions}) = \mathbf{0.07843}$
- $P(X > 3 \text{ fraudulent transactions}) = \mathbf{0.01776}$
- And so on.
- This gives banks an idea of how likely it is that more than a certain number of fraudulent transactions will occur in a given day.

Real Life Examples of the Binomial Distribution

- **Example 3: Number of Spam Emails per Day**
- Email companies use the binomial distribution to model the probability that a certain number of spam emails land in an inbox per day.
-

Real Life Examples of the Binomial Distribution

- For example, suppose it is known that 4% of all emails are spam. If an account receives 20 emails in a given day, we can use Binomial Probability to find the probability that a certain number of those emails are spam:
- $P(X = 0 \text{ spam emails}) = \mathbf{0.44200}$
- $P(X = 1 \text{ spam email}) = \mathbf{0.36834}$
- $P(X = 2 \text{ spam emails}) = \mathbf{0.14580}$

Real Life Examples of the Binomial Distribution

- **Example 4: Number of River Overflows**
- Park systems use the binomial distribution to model the probability that rivers overflow a certain number of times each year due to excessive rain.

Real Life Examples of the Binomial Distribution

- For example, suppose it is known that a given river overflows during 5% of all storms. If there are 20 storms in a given year, we can use the Binomial Probability to find the probability that the river overflows a certain number of times:
- $P(X = 0 \text{ overflows}) = \mathbf{0.35849}$
- $P(X = 1 \text{ overflow}) = \mathbf{0.37735}$
- $P(X = 2 \text{ overflows}) = \mathbf{0.18868}$

This gives the parks departments an idea of how many times they may need to prepare for overflows throughout the year.

Real Life Examples of the Binomial Distribution

- **Example 5: Shopping Returns per Week**
- Retail stores use the binomial distribution to model the probability that they receive a certain number of shopping returns each week.

Real Life Examples of the Binomial Distribution

- For example, suppose it is known that 10% of all orders get returned at a certain store each week. If there are 50 orders that week, we can use the Binomial Probability to find the probability that the store receives more than a certain number of returns that week:
- $P(X > 5 \text{ returns}) = \mathbf{0.18492}$
- $P(X > 10 \text{ returns}) = \mathbf{0.00935}$
- $P(X > 15 \text{ returns}) = \mathbf{0.00002}$
- This gives the store an idea of how many customer service reps they need to have in the store that week to handle returns.

Binomial Probability Applet Calculator

- <https://homepage.divms.uiowa.edu/~mbognar/applets/bin.html>